

Backpropagation multi-layer perceptron for incomplete pairwise comparison matrices in analytic hierarchy process

Yi-Chung Hu^{a,*}, Jung-Fa Tsai^b

^a Department of Business Administration, Chung Yuan Christian University, Chung-Li 320, Taiwan, ROC

^b Department of Business Management, National Taipei University of Technology, Taipei 106, Taiwan, ROC

Abstract

Analytic hierarchy process (AHP) is a widely used decision making method in many areas such as management sciences. The performance ratings of multiple criteria and alternatives can be elicited from pairwise comparisons obtained by expressing the decision maker's perceive. However, it may be difficult for the decision maker to prudently assign values to comparisons for large number of criteria or alternatives. Since there exist distinct relationships between any two elements in the real world, the relationship between the missing comparison and the assigned comparisons should be taken into account. The aim of this paper is to propose a novel method using a well-known regression tool, namely the backpropagation multi-layer perceptron (i.e., MLP) to realize the above implicit relationship so as to estimate a missing pairwise judgment from the other assigned entries. A computer simulation is employed to demonstrate that the proposed method can effectively find a missing entry of an incomplete pairwise matrix such that its consistency index is minimized. © 2005 Elsevier Inc. All rights reserved.

Keywords: Analytic hierarchy process; Neural network; Missing comparison; Function approximation; Decision making

1. Introduction

The well-known analytic hierarchy process (AHP), proposed by Saaty [1–3], has been a widely used decision making method in many areas such as management sciences [4,5]. The AHP decomposes a complex decision problem into a hierarchical structure. At a time, for criteria say c_1 and c_2 , the decision maker chooses one linguistic phrase, such as “ c_1 is more important than c_2 ” or “ c_1 and c_2 are of equal importance”, in order to make a pairwise comparison between c_1 and c_2 . Preference ratings of multiple criteria or alternatives can thus be determined by analyzing the pairwise comparison matrix [6]. The principal eigenvector, normalized by the sum of elements in the above eigenvector, of the matrix indicates relative importances of individual criteria or alternatives. For effectively finding the principal eigenvector, many methods have been proposed such as the power method [7], the geometric mean method [2], the optimization approach [8], the neural network-based model [9], and the human rationality assumption [10].

* Corresponding author.

E-mail address: ychu@cycu.edu.tw (Y.-C. Hu).

If there are n criteria, then the decision maker has to conduct $n(n-1)/2$ pairwise comparisons. It is clear that large numbers of comparisons would be made by the decision maker due to large n . For a respondent, he or she may become tired and inattentive with selecting linguistic phrases [10]. Hence, it is quite possible that missing comparisons would be generated in a pairwise comparison matrix with large n . Also, many effective methods have been proposed to estimate missing comparisons in an incomplete comparison matrix, such as connecting paths [10–12], the revised geometric mean method (RGM) [13], and the characteristic polynomial-based method [14,15]. In principle, these methods aim to minimize the consistent index (CI) of the incomplete matrix [14]. That is, the CI of a pairwise comparison matrix is desired to be as small as possible.

Actually, there exist distinct relationships between any two subsystems in the real world, although we do not know exactly what these relationships are [16,17]. This implies that a relationship between missing comparisons and the assigned comparisons in an incomplete pairwise comparison matrix should be taken into account. In other words, we consider that the assigned entries can be employed to estimate the missing entry. In this paper, the backpropagation multi-layer perceptron (MLP) [18,19], which is usually used as a tool of the approximation of functions like regression [20], are considered to realize the above relationship. For simplicity, only one missing comparison is taken into account. That is, the assigned entries in the upper-triangular region of an input matrix are actually used as the input of a multi-layer neural network, respectively. Then, the missing entry can be obtained from the actual output of the above network.

The aim of this paper is to propose a heuristic method by the above-mentioned MLP to estimate a missing entry from assigned entries in an incomplete comparison matrix. We further employ a computer simulation to demonstrate that the proposed method can effectively find a missing entry of an incomplete matrix such that its consistency index is minimized. That is, it seems that the proposed method can be used as an effective tool for recognizing the usefulness of an incomplete matrix.

The rest of this paper is organized as follows. The AHP is briefly introduced in Sections 2. In Section 3, the aforementioned estimation methods (i.e., connecting paths, RGM, and the characteristic polynomial-based method) are also briefly described. Subsequently, the proposed neural network-based method is presented in detail in Section 4. In Section 5, a computer simulation with two thousands randomly-generated matrices are used to examine the performance of the proposed method by comparing CI of the proposed method with those of the other methods introduced in Section 3. We end this paper with discussions and conclusions in Section 6.

2. Analytic hierarchy process (AHP)

The AHP is an approach based on pairwise comparisons [6], that can be used to determine relative weights of individual criteria or alternatives. Pairwise comparisons are usually quantified by the linear scale or the nine-point intensity scale proposed by Saaty [2]. By the linear scale, each linguistic phrase is mapped to one value in a set of available values, namely $\{9, 8, 7, 6, 5, 4, 3, 2, 1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9\}$. An $n \times n$ positive reciprocal matrix (i.e., pairwise comparison matrix), $\mathbf{A}$, is thus generated as follows:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \quad (1)$$

where $a_{ij} \in \{9, 8, 7, 6, 5, 4, 3, 2, 1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9\}$, and $a_{ii} = 1$ where $1 \leq i, j \leq n$. a_{ij} actually represents a pairwise judgment. In addition, the following equation holds:

$$a_{ij} = 1/a_{ji}. \quad (2)$$

If $\mathbf{A}$ is perfectly consistent, then the principal eigenvalue, $\lambda_{\max}$, is equal to n . That is,

$$\mathbf{A}\mathbf{w} = n\mathbf{w}, \quad (3)$$

where $\mathbf{w} = (w_1, w_2, \dots, w_n)^T$ is the principal eigenvector corresponding to $\lambda_{\max}$, and can be determined by an iterative procedure, namely the power method. Furthermore, the transitivity property, $a_{ij} \times a_{jk} = a_{ik}$, holds.

The relative weight of the criterion, c_i , or alternative, A_i , can be determined by normalizing w_i (i.e., $w_i / \sum_{j=1, \dots, n} w_j$) [6,21]. However, since it is difficult to satisfy the transitivity property during the evaluation process, $\mathbf{A}$ is usually inconsistent such that $\lambda_{\max}$ is greater than n . Saaty [1] defined the CI in order to determine the degree of inconsistency of $\mathbf{A}$:

$$CI = \frac{\lambda_{\max} - n}{n - 1}. \quad (4)$$

Saaty also suggested that CI should be below 0.1. In principle, the matrices with CI above 0.1 cannot be processed for further preference analysis, and may be treated as a useless matrix. The power method for finding the principal eigenvector of a positive reciprocal matrix is described as follows:

Algorithm (*The power method for finding the principal eigenvector*).

Input: A positive reciprocal matrix, $\mathbf{A}$

Output: The principal eigenvector of $\mathbf{A}$

Method:

Step 1. Initialization

Generate an arbitrary vector, $\mathbf{v} = (v_1, v_2, \dots, v_n)^T$. Also, set λ to a large number, say 100.

Step 2. Compute $\mathbf{A}\mathbf{v}$

Set $\mathbf{A}\mathbf{v} = \mathbf{u} = (u_1, u_2, \dots, u_n)^T$.

Step 3. Normalize $\mathbf{u}$

Normalize $\mathbf{u}$ such that $u_i = u_i / \sum_{j=1, \dots, n} u_j$.

Step 4. Termination test

If the absolute value of the difference between $\sum_{j=1, \dots, n} u_j$ and λ reaches above a tolerant error, say 0.001, then set $\lambda = \sum_{j=1, \dots, n} u_j$, $\mathbf{v} = \mathbf{u}$, and return to Step 2; otherwise, stop the whole procedure.

At last, $\mathbf{u}$ and $\sum_{j=1, \dots, n} u_j$ converge to $\mathbf{w}$ and $\lambda_{\max}$, respectively. The power method is also used in our method. The details of the AHP can be found in [2,6].

3. Missing comparisons

As we have mentioned above, a respondent may become tired and inattentive with selecting linguistic phrases for large numbers of pairwise comparisons. Sometimes, the decision maker may not make a direction evaluation of some comparisons [10]. An incomplete pairwise comparison matrix resulting from missing comparisons could be generated in the above situation.

Additionally, it is impossible to know whether an incomplete pairwise comparison matrix is consistent or not in advance. However, since the decision maker could try to minimize the error involved in pairwise comparisons from the human rationality assumption [10,22], an important task is to estimate the missing entry so as to minimize the CI of the above matrix. For this goal, it seems that the smaller CI obtained by an estimation method, the more effective this method is. It seems that several well-known estimation methods, such as the connecting paths [10–12], the RGM [13], and the characteristic polynomial-based method [14,15], aim to achieve the goal of minimizing the CI for an estimated matrix.

Since the estimation results of the proposed method can be compared with those of the aforementioned methods, they are briefly introduced in individual subsections below.

3.1. Connecting paths

Suppose that a_{ij} ($i \neq j$) is missed in a $n \times n$ positive reciprocal matrix. Then, a_{ij} can be determined by a connecting path of size k , cp_k , as follows:

$$cp_k = a_{i,p1} \times a_{p1,p2} \times \dots \times a_{pk,j}. \quad (5)$$

Actually, a_{ij} is equal to the geometric mean of all connecting paths related to a_{ij} . For instance, if there are three such connecting paths, such as cp_1 , cp_2 , and cp_3 , then

$$a_{ij} = \sqrt[3]{\prod_{t=1}^3 cp_t}. \quad (6)$$

However, a main drawback of connecting paths is that the number of connecting paths may be astronomically large [10]. For instance, the number of possible connecting paths is equal to 109,600 for a pairwise comparison matrix of dimension of 10.

3.2. Revised geometric mean method

RGM converts the incomplete pairwise comparison matrix, $\mathbf{A}$, into a transformed matrix, $\mathbf{C}$, rather than assigning an estimation value to the missing entry. Then, the principal eigenvector and $\lambda_{\max}$ of the above transformed matrix are determined. $\mathbf{C}$ is determined as follows:

$$c_{ii} = 1 + m_i, \quad \text{where } m_i \text{ is the number of missing comparisons in the } i\text{th row}, \quad (7)$$

$$c_{ij} = a_{ij}, \quad \text{if } i \neq j \text{ and } a_{ij} \text{ is not a missing comparison}, \quad (8)$$

$$c_{ij} = 0, \quad \text{if } i \neq j \text{ and } a_{ij} \text{ is a missing comparison}. \quad (9)$$

Of course, if a_{ij} is missed, then a_{ji} is also missed. For instance, an example of RGM was illustrated by Harker [13] as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & - \\ \frac{1}{2} & 1 & 2 \\ - & \frac{1}{2} & 1 \end{bmatrix}, \quad (10)$$

where “–” denotes a missing element. Then a transformed matrix $\mathbf{C}$ can be further determined as follows:

$$\mathbf{C} = \begin{bmatrix} 2 & 2 & 0 \\ \frac{1}{2} & 1 & 2 \\ 0 & \frac{1}{2} & 2 \end{bmatrix}. \quad (11)$$

That is, each missing element is replaced with zero. In general, if a_{ij} ($i \neq j$) is missed in a $n \times n$ positive reciprocal matrix, then both a_{ii} and a_{jj} can be replaced with 2, and both a_{ij} and a_{ji} can be replaced with zero.

3.3. Characteristic polynomial-based method

Let the characteristic polynomial of a positive reciprocal matrix, $\mathbf{A}$, of order n as follows:

$$P_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A}), \quad (12)$$

where $\mathbf{I}$ is a identity matrix. Shiraishi et al. [14] demonstrated that CI of $\mathbf{A}$ is equal to zero if and only if the coefficient of λ^{n-3} , say c_3 , of $P_{\mathbf{A}}(\lambda)$ is equal to zero. In other words, CI of $\mathbf{A}$ can be as small as possible by making c_3 approached to zero. Additionally, since c_3 is a negative value, “CI of $\mathbf{A}$ is smaller if and only if c_3 is larger” also holds.

Since c_3 is required to be as large as possible, the following maximum problem is taken into account:

$$\text{Max}_x c_3(x). \quad (13)$$

That is, find x , which can be assigned to the missing entry, say a_{ij} , of the above problem such that c_3 can be maximized. Then, the missing entry (i.e., a_{ij}) can be estimated by the optimal solution, x^* , as follows:

$$x^* = \sqrt{\frac{\sum_{\substack{k=1 \\ k \neq i,j}}^n a_{ik} a_{kj}}{\sum_{\substack{k=1 \\ k \neq i,j}}^n \frac{1}{a_{ik} a_{kj}}}}. \quad (14)$$

4. The proposed neural network-based method

In this section, a heuristic method on the basis of multi-layer neural networks is proposed. First, a three-layer neural network architecture employed to estimate a missing entry is demonstrated in Section 4.1. Regarding the estimation error between the real value and the estimated value obtained by the neural network, the detailed algorithm of the proposed method is described in Section 4.2.

4.1. Multi-layer perceptron

A multi-layer perceptron trained by the well-known backpropagation algorithm can be used as a tool of the approximation of functions like regression [20]. Actually, it has been proved that a single hidden layer is sufficient to approximate any continuous function [23]. A three-layer perceptron is thus taken into account in this paper. However, in general, the appropriate numbers of hidden nodes cannot be known in advance [18].

A regression tool can be used to realize a relationship between input attributes and output attributes. This paper considers that it is feasible for realizing an implicit relationship between missing comparisons and the assigned comparisons in an incomplete pairwise comparison matrix by a multi-layer neural network. It should be noted that, it may be not possible to directly implement the above mapping or relationship only by a linear function since the considered attributes are not always independent of each other [24].

For simplicity, only one missing comparison is taken into account. Thus, for a $n \times n$ pairwise comparison matrix, the missing comparison is equal to the actual output of a trained neural network, and the other $[n(n-1)/2 - 1]$ assigned comparisons in the upper-triangular region are taken as inputs. In other words, assigned comparisons are used to estimate the missing comparison by the neural network. For example, if a_{34} is missed in a 4×4 positive reciprocal matrix, then a_{34} can be obtained from the actual output of a trained neural network, and a_{12} , a_{13} , a_{14} , a_{23} , and a_{24} can be taken as the inputs. We illustrate the architecture of such a three-layer perceptron using the above-mentioned inputs and an output in Fig. 1. We can see that there are six input nodes (i.e., five nodes for a_{12} , a_{13} , a_{14} , a_{23} , and a_{24} , and one bias node with input 1) and one output node.

Using the back-propagation update rule, a three-layer perceptron is trained by adjusting its connection weights. In addition, the neural network converges to a set of weights as long as a termination condition or a stopping criterion is reached. Usually, the training procedure is terminated when the root mean square error [9], RMSE, reaches below the pre-specified tolerant error.

$$\text{RMSE} = \frac{1}{r} \sum_{j=1}^r |d_j - o_j|, \quad (15)$$

where d_j and o_j are the desired output and the actual output of the j th input training data, respectively, and r is the number of training data. Therefore, the purpose of the backpropagation algorithm is to minimize RMSE. Also, the trained network performs the function approximation. For instance, a function f that can map a vector $(a_{12}, a_{13}, a_{14}, a_{23}, a_{24})$ to a_{34} such that:

$$a_{34} = f(a_{12}, a_{13}, a_{14}, a_{23}, a_{24}). \quad (16)$$

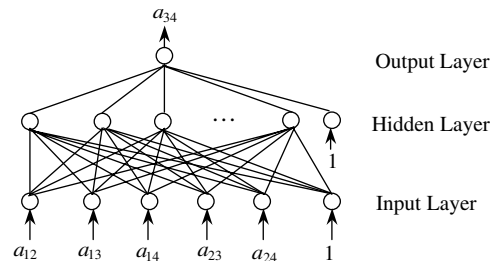


Fig. 1. A three-layer MLP.

As for the training data, the almost consistent matrices, whose consistent index approximate to zero, are actually taken into account. An almost perfectly consistent matrix is generated by the procedure described as follows:

Algorithm (Generation of an almost perfectly consistent pairwise comparison matrix).

Input: Matrix size, n .

Output: A $n \times n$ almost consistent matrix.

Method:

Step 1. Generate initial weights

Generate $w_1, w_2, \dots, w_n$ such that $\sum_{j=1, \dots, n} w_j = 1$.

Step 2. Generate matrix entries

2.1. Set a_{ii} to 1, for $i = 1 \dots n$;

2.2. a_{ij} is computed by approximate w_i/w_j to one number in the set of the Saaty scale, where $1 \leq i < j \leq n$. Subsequently, set a_{ji} by Eq. (2).

In Step 2, for instance, if $w_3/w_4 = 2.753$ in a $n \times n$ pairwise matrix, then a_{34} can be approximated by the value 3 since 3 is the closed value to 2.753 when the Saaty scale is used. This is similar to the construction of the Closest Discrete Pairwise matrix (CDP) proposed by Triantaphyllou [6].

4.2. Estimation error

Without losing generality, for a missing pairwise judgment, say $a'_{n-1,n}$, in the i th incomplete pairwise matrix, $\mathbf{A}^i$, there are unknown real values, say $\beta'_{n-1,n}$ and $\frac{1}{\beta'_{n-1,n}}$, that can be assigned to $a'_{n-1,n}$ and $a'_{n,n-1}$, respectively, so as to minimize the resulting matrix's CI. The trained neural network with respect to $n \times n$ matrices can assign an estimated value, $\beta'_{n-1,n}$, to $a'_{n-1,n}$. Also, $\frac{1}{\beta'_{n-1,n}}$ can be assigned to $a'_{n,n-1}$. When $\mathbf{A}^i$ is not a training data, there is an estimation error between $\beta'_{n-1,n}$ and $\beta_{n-1,n}^i$. It is reasonable to assume that there are two average errors, $\xi_{n-1,n}$ and $\xi_{n,n-1}$ for test data such that,

$$\beta_{n-1,n}^i = \beta'_{n-1,n} + \xi_{n-1,n}, \quad (17)$$

$$\frac{1}{\beta_{n-1,n}^i} = \frac{1}{\beta'_{n-1,n}} + \xi_{n,n-1}, \quad (18)$$

where $\xi_{n-1,n}$ and $\xi_{n,n-1}$ are real numbers. Then,

$$\mathbf{A}^i \mathbf{w}^i = \begin{bmatrix} 1 & a_{12}^i & \cdots & a_{1n}^i \\ a_{21}^i & 1 & \cdots & a_{2n}^i \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,1}^i & a_{n-1,2}^i & \cdots & \beta_{n-1,n}^i \\ a_{n1}^i & a_{n2}^i & \cdots & 1 \end{bmatrix} \begin{bmatrix} w_1^i \\ w_2^i \\ \vdots \\ w_n^i \end{bmatrix} \approx \begin{bmatrix} w_1^i + a_{12}^i w_2^i + \cdots + a_{1n}^i w_n^i \\ a_{21}^i w_1^i + w_2^i + \cdots + a_{2n}^i w_n^i \\ \vdots \\ a_{n-1,1}^i w_1^i + a_{n-1,2}^i w_2^i + \cdots + \beta'_{n-1,n} w_n^i + \xi_{n-1,n} w_n^i \\ a_{n1}^i w_1^i + a_{n2}^i w_2^i + \cdots + \frac{1}{\beta'_{n-1,n}} w_{n-1}^i + \xi_{n,n-1} w_{n-1}^i + w_n^i \end{bmatrix}. \quad (19)$$

Then, $\xi_{n-1,n} w_n^i$ and $\xi_{n,n-1} w_{n-1}^i$ are heuristically transformed into $\delta_{n-1,n} w_{n-1}^i$ and $\delta_{n,n-1} w_n^i$, respectively, where $\delta_{n-1,n}$ and $\delta_{n,n-1}$ are real numbers. In other words, $\xi_{n-1,n}$ and $\xi_{n,n-1}$ are assumed to be equal to $\delta_{n-1,n} w_{n-1}^i / w_n^i$ and $\delta_{n,n-1} w_n^i / w_{n-1}^i$, respectively. A matrix $\mathbf{B}^i$ can be further derived as follows:

$$\mathbf{B}^i = \begin{bmatrix} 1 & a_{12}^i & \cdots & a_{1n}^i \\ a_{21}^i & 1 & \cdots & a_{2n}^i \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,1}^i & a_{n-1,2}^i & \cdots & \beta_{n-1,n}^i \\ a_{n1}^i & a_{n2}^i & \cdots & 1 + \delta_{n,n-1} \end{bmatrix}. \quad (20)$$

Then, the performance ratings of criteria or alternatives can be determined as the principle eigenvector of $\mathbf{B}^i$, since $\mathbf{A}^i \mathbf{w}^i = \mathbf{B}^i \mathbf{w}^i$ holds. It is noted that $1 + \delta_{n-1,n}$ is positive or negative. This is also true for $1 + \delta_{n,n-1}$. The proposed method is described in details as follows.

Algorithm (*A neural network-based method for estimating a missing comparison*).

Input: (1) The k th pairwise comparison matrix, $\mathbf{A}^k$, with a missing entry a_{ij}^k , where $i < j$.
(2) Matrix size, n .

Output: A three-layer perceptron for estimating a_{ij}^k .

Method:

Step 1. Generate training data

r training data are generated by the algorithm for generating $n \times n$ almost perfectly consistent matrices.

Step 2. Train the neural network by the backpropagation algorithm

- Specify the learning rate and the momentum parameter in the back-propagation update rule. Both user-specified parameters are between 0 and 1. It would be helpful to employ the momentum parameter to achieve an objective of smooth convergence.
- Specify a threshold as the stopping criterion. The network reaches the convergence as long as the average RMSE below the above threshold.
- Use r training data to train the three-layer perceptron by the backpropagation algorithm with pattern-by-pattern mode updating. In this mode, a training data is presented at the input and then all connection weights are updated before the next training data is considered [9]. During the training process, a_{ij}^i corresponding to the i th matrix is used as the desired output, and the other upper-triangular elements are used as the inputs, where $i = 1, 2, \dots, r$.

Step 3. Specify the estimation error

Specify appropriate values to both $\delta_{n-1,n}$ and $\delta_{n,n-1}$.

Step 4. Estimation of missing entry

In addition to the missing entry in $\mathbf{A}^k$, the upper-triangular elements, a_{st}^k ($s < t$, and $1 \leq s, t \leq n$), are used as the inputs of the trained neural network. Thus, a_{ij}^k is equal to the actual output of the trained network. It is noted that $\mathbf{A}$ may be not an almost perfectly consistent matrix.

Step 5. Find the principal eigenvector

The power method can be used to find the principal eigenvector.

In our framework, $\mathbf{A}^k$ is a randomly-generated matrix rather than a training data. Thus, $\mathbf{A}^k$ may be not an almost perfectly consistent matrix. Of course, if the corresponding parameter specifications (i.e., connection weights, the architecture, and the matrix size) of the trained network have been prepared or stored once, then an input pairwise comparison matrix with a missing entry can be processed immediately without performing Steps 1 and 2 of the proposed method again. Below, a computer simulation is employed to demonstrate that the effectiveness of the proposed method in comparison with the other estimation methods.

5. Computer simulation

As we have mentioned above, for an incomplete pairwise matrix, the main aim of this paper is to minimize the consistency index by estimating the missing entry of the above matrix. Three-layer neural networks are trained for various sizes, such as 4×4 , 5×5 , 6×6 , 7×7 , 8×8 , and 9×9 , of pairwise comparison matrices.

By using the following parameter specifications, the proposed method is performed for matrices of order n :

- Size of training data: 1000 almost perfectly consistent matrices (i.e., $r = 1000$).
- Number of hidden nodes in the hidden layer: 50.
- In the back-propagation update rule, the learning rate and the momentum parameter are 0.5 and 0.2, respectively. It should be noted that the number of hidden nodes, the learning rate, and the momentum parameter are arbitrarily specified in order to train the backpropagation MLP.
- Specify a threshold, 0.01, as the stopping criterion.

- (e) Location of the missing entry: Without losing generality, the comparison in $(n - 1)$ th row and n th column is assumed to be missed.
- (f) Size of testing data: In order to examine the effectiveness and usefulness of the proposed method, 2000 $n \times n$ matrices are randomly generated, and the missing comparisons will be derived by the trained neural network. In addition to the missing entry, the upper-triangular elements of the above matrices are generated from uniform numbers in $1/9, 1/8, \dots, 1/2, 1, \dots, 9$. Then, the lower triangular elements are the reciprocal values of the upper-triangular elements.
- (g) $\delta_{n-1,n}$ and $\delta_{n,n-1}$: Actually, it is not easy to specify appropriate values to both $\delta_{n-1,n}$ and $\delta_{n,n-1}$. For simplicity, let $\delta_{n-1,n}$ be equal to $\delta_{n,n-1}$. Then, a real value, γ , which ranges from -0.1 to 0.1 , is set to $\delta_{n-1,n}$ and $\delta_{n,n-1}$, where $1 \leq k \leq 2000$.

The connecting paths, the RGM, and the characteristic polynomial-based method (CPB) are also applied to the testing data with the same location of a missing entry. Thus, four different CI of an incomplete matrix can be obtained by the aforementioned four estimation methods (i.e., the connecting paths, the RGM, the CPB, and the proposed method). Then, if an estimated matrix's CI obtained by one method is smallest, then this method is considered as a winner among the above four methods. It is found that poor results can be obtained by the proposed method if positive values related to $\delta_{n-1,n}$ are concerned. Therefore, regarding negative values related to $\delta_{n-1,n}$, the simulation results are demonstrated in Table 1.

Table 1 summarizes the number of becoming winners of individual methods under various parameter specifications of $\delta_{n-1,n}$ and the dimension of a comparison matrix. For example, for a matrix of dimension of 6, the number of becoming winners of the RGM is 668 when $\delta_{n-1,n}$ is equal to 0.01. In such a case, connecting paths performs best since the number of becoming winners is 730. From Table 1, some results are summarized as follows:

Table 1
The number of winners of various methods

Dimension	Method	$\delta_{n-1,n}$										
		0.000	-0.005	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
4×4	RGM	47	31	38	37	32	32	31	29	32	26	30
	CPB	15	3	7	6	7	4	8	4	6	5	3
	Connecting paths	1924	1662	1543	1403	1295	1222	1139	1099	1033	999	954
	The proposed method	14	304	412	555	666	742	822	868	929	970	1013
5×5	RGM	510	474	462	443	434	419	403	393	384	372	364
	CPB	167	160	157	148	144	144	142	145	139	139	136
	Connecting paths	1017	974	944	898	861	828	805	785	763	748	732
	The proposed method	306	392	437	511	561	609	650	677	714	741	768
6×6	RGM	710	682	668	628	599	586	573	558	536	521	513
	CPB	204	191	191	185	180	170	158	151	148	143	140
	Connecting paths	813	758	730	699	671	641	605	582	568	550	525
	The proposed method	273	369	411	488	550	603	664	709	748	786	822
7×7	RGM	718	674	642	577	534	500	473	449	413	393	364
	CPB	219	198	189	187	182	170	160	147	141	140	134
	Connecting paths	642	598	573	532	500	477	456	435	411	389	379
	The proposed method	421	530	596	704	784	853	911	969	1035	1078	1123
8×8	RGM	708	592	544	455	387	339	297	276	263	232	218
	CPB	213	169	163	148	135	123	110	95	78	70	62
	Connecting paths	565	515	470	406	367	328	303	273	256	238	221
	The proposed method	514	724	823	991	1111	1210	1290	1356	1403	1460	1499
9×9	RGM	756	666	607	541	472	407	374	336	294	270	246
	CPB	233	209	185	175	154	139	121	109	102	93	83
	Connecting paths	529	455	435	377	343	308	284	257	239	217	203
	The proposed method	482	670	773	907	1031	1146	1221	1298	1365	1420	1468

- (a) For a matrix of any dimension, it is clear that the smaller value of $\delta_{n-1,n}$, the larger number of becoming winners can be obtained by the proposed method. For instance, for a matrix of dimension of 8, the number of becoming winners corresponding to -0.02 (i.e., 991) is larger than that corresponding to -0.005 (i.e., 724).
- (b) For a larger dimension of a matrix, the proposed method can outperform the other estimation method when smaller $\delta_{n-1,n}$ is pre-specified. For instance, the proposed method performs best when the dimension is 6 and $\delta_{n-1,n}$ is at least below -0.05 , and when the dimension is 7 and $\delta_{n-1,n}$ is at least below -0.01 , and when the dimension is 8 and $\delta_{n-1,n}$ is at least below -0.005 . It seems that a smaller value of $\delta_{n-1,n}$ (e.g., -0.06 and -0.08) is unnecessary when the dimension of a pairwise comparison matrix is larger (e.g., 7 and 8).

By using the backpropagation multi-layer perceptron to implement a relationship between the missing comparison and the assigned comparisons in an incomplete pairwise comparison matrix, the proposed method, using a larger value of $\delta_{n-1,n}$, performs well in comparison with the other estimation methods when the order n is larger and $\delta_{n-1,n}$ is negative.

6. Discussion and conclusions

In this paper, a missing entry in the incomplete pairwise comparison matrix is estimated by a three-layer neural network. In particular, from the computer simulation demonstrated in the previous section, we can see that the estimation performance can be greatly improved by considering the difference between the real value and the estimated value (i.e., $\xi_{n-1,n}$). In addition, estimation results of the proposed method with appropriate parameter setting (i.e., $\delta_{n-1,n}$) can outperform those of the other three estimation methods. It seems that the proposed method can perform quite well in the case of larger order (e.g., $n = 8, 9$).

In comparison with the proposed method, it is obvious that other methods in this paper don't require training samples, and can give results in due course. However, we consider that the re-training cannot be often performed as the training task has been performed once. Also, In comparison with the effectiveness, the efficiency may be not an important factor for several applications. Additionally, only one missing element is taken into account for simplicity. Of course, many entries may be missed in a questionnaire. However, if there are too many missing entries in a pairwise comparison matrix, then such a questionnaire should be discarded. Hence, it is worth to discuss the possible upper bound of missing entries in a positive reciprocal matrix in the future.

In addition to the three-layer neural network, there are several optimization tools related to soft computing could be also employed to estimate the missing entry, such as the fuzzy rule-based systems. Actually, the fuzzy rule-based systems have been widely applied to function approximation problems [25,26]. The main advantage of fuzzy rule-based systems is to provide linguistic interpretations. Also, it was proven that the well-known zero-order Sugeno fuzzy model [27] can approximate any nonlinear function on a compact set to an arbitrary degree of accuracy under certain conditions [28]. Hence, the application of the fuzzy rule-based systems to estimate the missing entry remains as the future study.

Acknowledgements

The authors would like to thank the anonymous referees for taking their valuable time to help them on this paper.

References

- [1] T.L. Saaty, A scaling method for priorities in hierarchical structure, *Journal of Mathematical Psychology* 15 (1977) 237–281.
- [2] T.L. Saaty, *The Analytic Hierarchy Process*, McGraw-Hill, New York, 1980.
- [3] T.L. Saaty, *Fundamentals of Decision Making and Priority Theory with The Analytic Hierarchy Process*, vol. VI, RWS Publication, Pittsburgh, 1994.
- [4] K.M. Al-Harbi Al-Subhi, Application of the AHP in project management, *International Journal of Project Management* 19 (1) (2001) 19–27.

- [5] S.H. Tsaur, T.Y. Chang, C.H. Yen, The evaluation of airline service quality by fuzzy MCDM, *Tourism Management* 23 (2002) 107–115.
- [6] E. Triantaphyllou, *Multi-Criteria Decision Making Methods: A Comparative Study*, Kluwer, Boston, MA, 2000.
- [7] R.L. Burden, J.D. Faires, *Numerical Analysis*, fifth ed., PWS-Kent, Boston, MA, 1993.
- [8] A.T.W. Chu, R.E. Kalaba, K. Spingarn, A comparison of two methods for determining the weights of belonging to fuzzy sets, *Journal of Optimization Theory and Application* 27 (4) (1979) 531–538.
- [9] A. Stam, M. Sun, M. Haines, Artificial neural network representations for hierarchical preference structures, *Computers and Operations Research* 23 (12) (1996) 1191–1201.
- [10] Q. Chen, E. Triantaphyllou, Estimating data for multi-criteria decision making problems: Optimization techniques, in: P.M. Pardalos, C. Floudas (Eds.), *Encyclopedia of Optimization*, vol. 2, Kluwer, Boston, MA, 2001, pp. 27–36.
- [11] P.T. Harker, Incomplete pairwise comparisons in the analytic hierarchy process, *Mathematic Modeling* 9 (11) (1987) 837–848.
- [12] P.T. Harker, Derivatives of the Perron root of a positive reciprocal matrix: with application to the analytic hierarchy process, *Applied Mathematic and Computation* 22 (1987) 217–232.
- [13] P.T. Harker, Alternative modes of questioning in the analytic hierarchy process, *Mathematic Modeling* 9 (3–5) (1987) 353–360.
- [14] S. Shiraishi, T. Obata, M. Daigo, Properties of a positive reciprocal matrix and their application to AHP, *Journal of the Operations Research Society of Japan* 41 (1998) 404–414.
- [15] T. Obata, S. Shiraishi, M. Daigo, N. Nakajima, Assessment for an incomplete comparison matrix and improvement of inconsistency: computational experiments, in: *Proceedings of the 5th International Symposium on the Analytic Hierarchy Process*, Kobe, Japan, 1999, pp. 200–205.
- [16] J.L. Deng, Control problems of grey systems, *Systems and Control Letters* 1 (5) (1982) 288–294.
- [17] Y.C. Hu, R.S. Chen, Y.T. Hsu, G.H. Tzeng, Grey self-organizing feature maps, *Neurocomputing* 48 (1) (2002) 863–877.
- [18] J. Hertz, A. Krogh, R.G. Palmer, *Introduction to The Theory of Neural Computation*, Addison-Wesley, CA, 1991.
- [19] M.H. Hassoun, *Fundamentals of Artificial Neural Networks*, MIT Press, Cambridge, 1995.
- [20] K.A. Smith, J.N.D. Gupta, Neural networks in business: techniques and applications for the operations researcher, *Computers and Operations Research* 27 (11–12) (2000) 1023–1044.
- [21] K.P. Yoon, C.L. Hwang, *Multiple Attribute Decision Making: An Introduction*, Sage Publications, London, 1995.
- [22] E.P. Triantaphyllou, M. Pardalos, S.H. Mann, A minimization approach to membership evaluation in fuzzy sets and error analysis, *Journal of Optimization Theory and Application* 66 (2) (1990) 275–287.
- [23] G. Cybenko, Approximation by superpositions of a sigmoidal function, *Mathematics of Control, Signals, and Systems* 2 (1989) 303–314.
- [24] T.Y. Chen, H.L. Chang, G.H. Tzeng, Using fuzzy measures and habitual domains to analyze the public attitude and apply to the gas taxi policy, *European Journal of Operational Research* 137 (1) (2002) 145–161.
- [25] J.S.R. Jang, C.T. Sun, Neuro-fuzzy modeling and control, *Proceedings of the IEEE* 83 (3) (1995) 378–406.
- [26] K. Nozaki, H. Ishibuchi, H. Tanaka, A simple but powerful heuristic method for generating fuzzy rules from numerical data, *Fuzzy Sets and Systems* 86 (3) (1997) 251–270.
- [27] M. Sugeno, G.T. Kang, Structure identification of fuzzy control, *Fuzzy Sets and Systems* 28 (1988) 15–33.
- [28] J.S.R. Jang, ANFIS: adaptive-network-based fuzzy inference systems, *IEEE Transactions on Systems, Man, and Cybernetics* 23 (3) (1993) 665–685.